Leptomeningeal disease — evidence-first pipeline report

Run date: 2026-10-15 (ET). Stages A-C complete; Stage D (KERMT ADMET) status noted below.

1. What this is, in plain words

Leptomeningeal disease (LMD) is what happens when cancer spreads into the thin, fluid-filled lining that wraps the brain and spinal cord. Cancer cells float in the cerebrospinal fluid rather than sitting in one lump, so they are hard to reach with drugs and hard to measure. It is uncommon, it is serious, and it is badly under-studied.

What this run did. First it pulled every leptomeningeal study from ClinicalTrials.gov (219 found) and kept the **19 that have actual results posted** — those are the only ones with real patient outcome numbers to stand on. Then, for the drugs in those trials that are small molecules, it found the protein each one acts on and a real experimental crystal structure of that protein, generated close chemical relatives ("cousins") with MolMIM, docked everything with DiffDock, scored binding twice with Boltz-2, and finally ran the owner's locally trained KERMT model to predict absorption and safety properties.

What this cannot show. Nothing here is evidence that any drug works. The trial numbers come from small, uncontrolled studies. The model numbers are computer guesses about chemistry, not about patients. No new treatment is being proposed; this is a map of what is known plus hypotheses to test in a lab.

2. Trial evidence — the 19 studies with posted results

NCT	Study	Phase	Drug(s)	Design	Enrolled	Headline posted outcome	Serious AEs / deaths	Modality
NCT02336451	A Phase II Study to Evaluate the Efficacy and Safety of Oral	P2	ceritinib	5 arms, non-randomized, no comparator (allocation NON_RANDOMIZED, model PARALLEL)	156	ORR by arm 35.7 / 30.0 / 50.0 / 59.1%; leptomeningeal arm (Arm 5) 16.7%	serious AEs 150; deaths 62	SMALL MOLECULE (ceritinib)
NCT00445965	Iodine I 131 Monoclonal Antibody 3F8 in Treating Patients Wi	P2	—	single-arm (allocation NA, model SINGLE_GROUP)	78	67/78 alive at 6 months; 31 complete responses, 8 stable, 21 progressed, 18 not evaluable	serious AEs 39; deaths 54	BIOLOGIC (radiolabeled antibody 131I-3F8)
NCT00450866	Epothilone B in Treating Patients With CNS Metastases From B	P2	patupilone	single-arm (allocation NA, model SINGLE_GROUP)	55	CNS progression-free at 3 months: 12 participants (Group A), 2 (Group B)	serious AEs 18; deaths 0+ (some groups not posted)	SMALL MOLECULE (patupilone/epothilone B; record labels it "Biological")
NCT03275402	131I-omburtamab Radioimmunotherapy for Neuroblastoma Central	P2, P3	—	single-arm (allocation NA, model SINGLE_GROUP)	52	3-year overall survival rate 0.65	serious AEs 20; deaths 17	BIOLOGIC (radiolabeled antibody 131I-omburtamab)
NCT04420598	DS-8201a for trEatment of aBc, BRain Mets, And Her2[+] Disea	P2	—	single-arm (allocation NA, model SINGLE_GROUP)	41	Cohort 1 16-week PFS: 1 of 8; ORR cohorts 2/3/4: 7, 5, 3 responders	serious AEs 11; deaths 22	BIOLOGIC (antibody-drug conjugate trastuzumab deruxtecan)
NCT00801216	High-dose Sequential Chemoimmunotherapy for B-cell Lymphomas	P2	methotrexate, cytarabine, cyclophosphamide, etoposide, thiotepa, carmustine	single-arm (allocation NA, model SINGLE_GROUP)	40	2-year event-free survival: 16 participants (of 40 enrolled, 38 at risk)	serious AEs 4; deaths 22	MIXED regimen: small molecules (methotrexate, cytarabine, cyclophosphamide, etoposide, thiotepa, carmustine) + rituximab (antibody) + stem-cell transplant + radiation
NCT02616393	Phase 2 Study of Study of Tesevatinib in Subjects With NSCLC	P2	tesevatinib	3 arms, non-randomized, no comparator (allocation NON_RANDOMIZED, model SINGLE_GROUP)	36	Brain-met ORR 31.3% overall; leptomeningeal cohort (B) ORR 0%	serious AEs 42; deaths 36	SMALL MOLECULE (tesevatinib)

NCT	Study	Phase	Drug(s)	Design	Enrolled	Headline posted outcome	Serious AEs / deaths	Modality
NCT01325207	Intrathecal Trastuzumab for Leptomeningeal Metastases in HER	P1, P2	—	single-arm (allocation not stated, model SEQUENTIAL)	34	Best response: 6 partial responses, 18 stable, 0 complete; 1 dose-limiting toxicity at 80 mg	serious AEs 21; deaths 30	BIOLOGIC (intrathecal trastuzumab antibody)
NCT04250545	Testing of the Anti Cancer Drugs CB-839 HCl (Telaglenastat)	P1	sapanisertib, telaglenastat	single-arm (allocation NA, model SINGLE_GROUP)	22	MTD/RP2D: NA (not established); overall responses 1 + 0 + 1 participants across dose levels	serious AEs 12; deaths 15	SMALL MOLECULE (sapanisertib + telaglenastat)
NCT00924820	A Pilot Study of Bevacizumab for Neoplastic Meningitis	P1, P2	—	single-arm (allocation NA, model SINGLE_GROUP)	20	CSF response rate 93% of participants; median time to progression 3 weeks	serious AEs 11; deaths 0+ (some groups not posted)	BIOLOGIC (bevacizumab antibody)
NCT02939300	Ipilimumab and Nivolumab in Leptomeningeal Metastases	P2	—	single-arm (allocation NA, model SINGLE_GROUP)	18	44% of participants alive at 3 months	serious AEs 10; deaths 13	BIOLOGIC (nivolumab + ipilimumab antibodies)
NCT03501979	Tucatinib, Trastuzumab, and Capecitabine for the Treatment o	P2	tucatinib, capecitabine	single-arm (allocation NA, model SINGLE_GROUP)	17	Survival after starting treatment: 6 participants (of 17)	serious AEs 1; deaths 11	MIXED: small molecules tucatinib + capecitabine, plus trastuzumab antibody
NCT02948751	Microfluidic Device to Diagnose Leptomeningeal Metastasis in	not stated	—	single-arm (allocation NA, model SINGLE_GROUP)	14	Diagnostic device: 3 participants with LM detected (of 14); no treatment effect measured	serious AEs 0; deaths 0	DEVICE/diagnostic (OncoCEE CSF test) — no treatment
NCT03091478	Pembrolizumab in Patients With Leptomeningeal Disease	P2	—	single-arm (allocation NA, model SINGLE_GROUP)	13	At 12 weeks: 2 complete, 1 partial, 2 stable, 8 progressive	serious AEs 7; deaths 9	BIOLOGIC (pembrolizumab antibody)
NCT00523939	DepoCyt for Active Lymphomatous or Leukemic Meningitis	P2	cytarabine	2 arms, non-randomized, no comparator (allocation NON_RANDOMIZED, model PARALLEL)	4	Response Rate: no value posted in the results module	serious AEs 3; deaths 0+ (some groups not posted)	SMALL MOLECULE (liposomal cytarabine, intrathecal)
NCT01596127	Intrathecal Rituximab in Lymphoid Malignancies Involving Cen	P1, P2	—	single-arm (allocation NA, model SINGLE_GROUP)	4	Response rate 0% at 2 weeks	serious AEs 1; deaths 1	BIOLOGIC (intrathecal rituximab antibody)
NCT04460937	Testing the Addition of an Anti-cancer Drug, Adavosertib, to	P1	adavosertib	single-arm (allocation NA, model SINGLE_GROUP)	4	Maximum tolerated dose 200 mg adavosertib with radiation	serious AEs 0; deaths 4	SMALL MOLECULE (adavosertib) + RADIATION
NCT00992602	Liposomal Cytarabine and High-Dose Methotrexate in Treating	P2	methotrexate, cytarabine	single-arm (allocation NA, model SINGLE_GROUP)	3	Survival free of neurological progression: 6 weeks	serious AEs 0; deaths 3	SMALL MOLECULE (methotrexate + liposomal cytarabine)
NCT00964743	Depocyt® With Sorafenib in Neoplastic Meningitis	NA	cytarabine, sorafenib	single-arm (allocation NA, model SINGLE_GROUP)	2	Safety-only readout: adverse events in 2 of 2 participants	serious AEs 2; deaths 0+ (some groups not posted)	SMALL MOLECULE (liposomal cytarabine + sorafenib)

3. Read this before the table above

None of the 19 studies had a control arm. Every one is single-arm or several non-randomised cohorts of the same drug. That means the posted outcomes describe how these particular patients did — they do **not** show that any drug beat another option, or beat no treatment at all.
Most studies were tiny. Enrollments include 2, 3, 4, 4, 4, 13, 14, 17, 18, 20 and 22 participants. At those sizes a single patient moves the headline number by tens of percent. Only NCT02336451 (ceritinib, n=156) is large enough for its response rates to mean much, and even there the leptomeningeal arm was 16.7%.
Deaths in these tables are mostly from the cancer itself, not from the drug; these are late-stage patients. Do not read a death count as a drug safety signal without the full record.

4. Molecule hypotheses — structure, docking, affinity, ADMET

Boltz-2 `affinity_pred_value` is log-scale: **lower = stronger predicted binding**. `binding_probability` is the model's own 0-1 chance the pair binds at all. DiffDock confidence is pose plausibility, not affinity.

Drug	Target	PDB (res. Å)	Bound ligand	DiffDock drug	DiffDock best cousin	Boltz-2 drug aff r1/r2	drug prob r1/r2	cousin aff r1/r2	cousin prob r1/r2	Rank stable	Evidence note
sapanisertib	mTOR kinase	4JSV (3.5000)	none (ADP; drug not bound). Drug-bound only in PI3Kα 6GVF (2.5 Å, ligand FE5)	0.3752	0.3059 (4)	0.2852 / -0.2500	0.4688 / 0.4375	0.2871 / 0.0703	0.3828 / 0.3438	yes	weaker — drug not present in the structure
telaglenastat	Glutaminase, kidney isoform (GLS)	5HL1 (2.4)	63J (CB-839)	-2.3377	-0.8458 (1)	-0.4707 / 0.2891	0.5078 / 0.5312	0.6484 / 1.1875	0.1465 / 0.1318	yes	drug co-crystallized
adavosertib	WEE1 kinase	5V5Y (1.9)	8X7 (MK-1775)	0.5123	-0.3001 (4)	-2.1250 / -2.2188	0.8594 / 0.8750	1.1875 / 1.2031	0.1934 / 0.1196	yes	drug co-crystallized
tucatinib	HER2 (ERBB2)	8VB5 (1.48)	A1AAC (HER2 exon20-insertion mutant)	-0.0764	-0.0856 (3)	0.3477 / 0.2754	0.4453 / 0.4258	1.3438 / 1.1797	0.0674 / 0.0449	yes	drug co-crystallized
capecitabine	Thymidylate synthase (via its metabolite FdUMP; prodrug)	6QXG (2.08)	UFP (FdUMP, the active metabolite — not capecitabine)	-0.6346	-0.5376 (4)	0.4453 / 0.5273	0.4062 / 0.3789	1.3984 / 1.2656	0.2334 / 0.3008	yes	weaker — drug not present in the structure
methotrexate	Dihydrofolate reductase (DHFR)	1U72 (1.9)	MTX (+NADPH); Leu22-variant study — check chain is wild-type	-0.0046	-0.4453 (4)	-2.0625 / -2.2188	0.9922 / 0.9922	1.7031 / 1.7109	0.1006 / 0.0923	yes	drug co-crystallized
ceritinib	ALK kinase	4MKC (2.01)	4MK (LDK378 = ceritinib)	0.1825	-0.1101 (1)	-1.4062 / -1.3516	0.8281 / 0.8281	0.8906 / 0.9844	0.0869 / 0.0986	yes	drug co-crystallized
tesevatinib	EGFR (also HER2, EPHB4)	1XKK (2.4)	none for tesevatinib; FMM = lapatinib, a quinazoline analog	-0.1589	0.2306 (4)	-1.6797 / -1.7500	0.7188 / 0.7227	1.6406 / 1.6250	0.0601 / 0.0588	yes	weaker — drug not present in the structure
sorafenib	Multi-kinase; BRAF/CRAF, VEGFR1-3, PDGFR-β, KIT, FLT3, RET	1UWH (2.95)	BAX (sorafenib, wild-type BRAF)	0.4197	0.3038 (5)	-1.0000 / -0.9766	0.5586 / 0.5508	1.3359 / 1.4531	0.4922 / 0.5156	yes	drug co-crystallized
etoposide	Topoisomerase II alpha	5GWK (3.152)	EVP (with DNA)	-3.8492	-1.4638 (3)	0.4277 / 0.4551	0.6641 / 0.6562	0.6523 / 0.2520	0.5547 / 0.5586	no	drug co-crystallized; **rank flipped between repeats — no ordering claimed**; real target is protein+DNA, only protein modelled
patupilone	Tubulin / microtubules (stabilizer)	7DAE (2.394)	EPB (epothilone B)	-0.1532	-0.1351 (2)	1.1016 / 0.9766	0.5547 / 0.4453	1.2578 / 1.2344	0.1309 / 0.1099	yes	drug co-crystallized
cytarabine	DNA polymerase (label: "appears to"; no specific protein named)	none chosen (—)	not applicable — no named protein target	missing	missing (—)	missing / missing	missing / missing	missing / missing	missing / missing	missing	not eligible for the molecule models — not applicable — no named protein target
cyclophosphamide	DNA (alkylator) — no protein target	not applicable (—)	—	missing	missing (—)	missing / missing	missing / missing	missing / missing	missing / missing	missing	not eligible for the molecule models — —
thiotepa	DNA (alkylator) — no protein target	not applicable (—)	—	missing	missing (—)	missing / missing	missing / missing	missing / missing	missing / missing	missing	not eligible for the molecule models — —

Drug	Target	PDB (res. Å)	Bound ligand	DiffDock drug	DiffDock best cousin	Boltz-2 drug aff r1/r2	drug prob r1/r2	cousin aff r1/r2	cousin prob r1/r2	Rank stable	Evidence note
carmustine	DNA/RNA alkylator — no protein target	not applicable (—)	—	missing	missing (—)	missing / missing	missing / missing	missing / missing	missing / missing	missing	not eligible for the molecule models — —

Two candidates failed DiffDock outright and are reported as failed, not substituted: `telaglenastat_cousin_2` and `tesevatinib_cousin_2` (service error: "Fail to read ligand molecule description").

KERMT ADMET (Stage D)

Checkpoint: v11 epoch-31, val MAE 0.39627. Values are in the model's own units (several are log-scale), rounded to 3 decimals.

Molecule	role	CL_microsome_human	CL_microsome_mouse	CL_microsome_rat	CL_total_dog	CL_total_human	CL_total_monkey	CL_total_rat	CYP2C8_inhibition	CYP2C9_inhibition	CYP2D6_inhibition	CYP3A4_inhibition	Dog_fraction_unbound_plasma	Human_fraction_unbound_plasma	Monkey_fraction_unbound_plasma	Rat_fraction_unbound_plasma	Papp_Caco2	Pgp_human	hERG_binding	LogD_pH_7.4	kinetic_logSaq	thermo_logSaq	VDss_dog	VDss_human	VDss_monkey	VDss_rat
telaglenastat	drug	4.717	4.724	4.665	0.907	0.390	1.108	0.885	3.413	3.099	3.815	3.542	-1.353	-2.191	-1.164	-1.834	0.560	0.610	3.352	3.560	-5.470	-5.581	0.246	-0.075	0.008	0.170
telaglenastat_cousin_1	cousin	3.980	4.596	4.354	1.188	0.756	0.977	1.442	4.089	4.102	3.942	4.206	-0.314	-0.556	-0.483	-0.465	1.260	0.558	4.507	1.857	-3.884	-2.304	0.094	0.071	-0.034	0.343
adavosertib	drug	4.148	4.290	4.456	0.969	0.728	0.909	1.435	4.028	3.889	4.176	3.985	-0.754	-0.931	-0.745	-0.778	0.855	0.616	3.977	2.371	-4.052	-3.436	0.232	0.162	0.053	0.341
adavosertib_cousin_4	cousin	4.136	4.927	4.714	1.162	0.779	1.116	1.378	3.611	3.690	3.407	3.845	-0.935	-1.392	-0.691	-1.213	1.213	0.370	3.296	3.062	-4.666	-4.315	0.298	0.179	0.150	0.609
tucatinib	drug	4.468	4.700	4.642	1.125	0.745	1.032	1.205	3.627	3.346	3.510	3.052	-0.977	-1.432	-0.835	-1.262	0.910	0.444	3.955	2.890	-4.746	-4.087	0.375	0.265	0.116	0.381
tucatinib_cousin_3	cousin	4.124	4.413	4.456	1.279	0.772	1.110	1.704	4.157	4.279	3.838	4.083	-0.374	-0.420	-0.375	-0.270	0.555	0.634	4.277	0.794	-3.872	-2.569	0.206	0.176	-0.010	0.464
ceritinib	drug	4.411	4.355	4.422	0.918	0.528	1.090	1.217	3.485	3.634	3.902	3.605	-1.227	-1.667	-1.016	-1.541	0.312	0.811	3.876	2.677	-5.131	-4.911	0.423	0.270	0.095	0.377
ceritinib_cousin_1	cousin	3.942	4.442	4.481	1.213	0.848	1.064	1.599	4.091	4.101	3.651	4.075	-0.446	-0.857	-0.469	-0.501	0.954	0.418	3.734	1.660	-3.964	-2.446	0.337	0.371	0.084	0.516
sorafenib	drug	4.352	4.515	4.604	0.915	0.446	1.220	0.879	3.109	3.056	3.469	3.316	-1.330	-1.784	-1.143	-1.669	0.557	0.643	3.524	3.294	-5.708	-5.616	0.215	-0.074	0.142	0.182
sorafenib_cousin_5	cousin	4.610	5.059	5.166	0.962	0.534	1.480	1.145	3.355	3.468	3.726	3.702	-1.324	-1.726	-1.108	-1.721	1.134	-0.067	3.254	4.277	-6.021	-6.348	0.281	0.125	0.302	0.385
methotrexate	drug	3.495	3.932	4.144	0.419	0.162	0.273	0.706	4.596	4.368	4.815	4.694	-0.486	-0.921	-0.406	-0.801	-0.509	0.488	5.683	-1.533	-3.832	-3.435	-0.365	-0.572	-0.566	-0.378
methotrexate_cousin_4	cousin	4.158	4.304	4.448	1.001	0.652	0.954	1.317	4.076	4.043	4.051	4.002	-0.405	-0.702	-0.523	-0.391	0.909	0.626	4.031	0.935	-3.833	-1.734	0.093	0.066	0.028	0.235
etoposide	drug	4.383	4.614	4.739	1.058	0.185	0.844	1.430	4.094	4.257	4.637	3.894	-1.006	-1.443	-1.032	-1.126	0.109	0.295	4.334	1.569	-4.026	-3.812	0.347	-0.046	-0.017	0.337
etoposide_cousin_3	cousin	3.990	4.582	4.577	1.291	0.940	1.072	1.610	4.195	4.364	3.888	4.153	-0.530	-0.812	-0.563	-0.499	0.864	0.087	4.236	1.018	-3.956	-2.585	0.460	0.378	0.194	0.589

Molecule	role	CL_microsome_human	CL_microsome_mouse	CL_microsome_rat	CL_total_dog	CL_total_human	CL_total_monkey	CL_total_rat	CYP2C8_inhibition	CYP2C9_inhibition	CYP2D6_inhibition	CYP3A4_inhibition	Dog_fraction_unbound_plasma	Human_fraction_unbound_plasma	Monkey_fraction_unbound_plasma	Rat_fraction_unbound_plasma	Papp_Caco2	Pgp_human	hERG_binding	LogD_pH_7.4	kinetic_logSaq	thermo_logSaq	VDss_dog	VDss_human	VDss_monkey	VDss_rat
patupilone	drug	4.807	4.918	5.165	1.344	0.583	0.876	1.462	4.094	4.303	4.238	3.460	-1.007	-0.838	-1.032	-1.059	1.303	0.263	4.387	2.856	-3.934	-2.926	0.437	0.307	0.185	0.667
patupilone_cousin_2	cousin	4.034	4.766	4.845	1.228	0.656	1.073	1.586	4.182	4.100	3.978	4.014	-0.571	-0.623	-0.643	-0.402	1.288	0.246	3.948	2.421	-3.888	-2.831	0.399	0.142	0.035	0.388
sapanisertib	drug	4.180	4.506	4.749	0.973	0.764	0.889	1.294	3.748	3.965	4.055	3.744	-0.370	-0.671	-0.688	-0.713	1.198	0.092	4.171	2.293	-4.087	-3.425	0.291	0.283	0.160	0.290
sapanisertib_cousin_4	cousin	4.035	4.444	4.644	1.085	0.886	1.020	1.605	3.973	4.188	4.047	3.978	-0.356	-0.544	-0.540	-0.352	0.738	0.327	3.831	1.145	-3.907	-2.887	0.172	0.385	0.055	0.420
tesevatinib	drug	4.539	4.785	4.799	1.058	0.674	1.256	1.581	3.685	3.807	3.655	3.746	-1.115	-1.521	-0.872	-1.339	0.997	0.251	3.402	3.581	-4.907	-4.716	0.463	0.459	0.206	0.609
tesevatinib_cousin_4	cousin	4.060	4.345	4.505	0.991	0.572	0.968	1.384	4.126	4.010	4.497	4.343	-0.397	-0.388	-0.510	-0.318	0.971	0.382	4.451	1.032	-3.925	-1.655	0.095	-0.044	-0.040	0.255
capecitabine	drug	3.964	4.329	4.312	0.845	0.514	0.810	1.177	4.059	4.225	4.419	4.499	-0.619	-0.868	-0.616	-0.700	0.816	0.398	4.441	1.142	-3.866	-3.099	-0.088	-0.110	-0.057	0.087
capecitabine_cousin_4	cousin	3.982	4.267	4.381	1.283	0.787	1.131	1.500	4.103	4.179	3.654	4.145	-0.294	-0.358	-0.330	-0.328	0.751	0.591	4.446	0.730	-3.945	-1.671	0.218	0.236	0.186	0.504

5. Honest readout

Most coherent across trial evidence + docking + affinity:

- **Ceritinib / ALK (NCT02336451).** The only study in the set with real size (n=156) and real response rates, and the chemistry agrees: ceritinib docks into its own crystal structure (4MKC), and Boltz-2 gives it the strongest numbers in the set after methotrexate (affinity -1.406 / -1.352, binding probability 0.828 in both repeats), beating its cousin in both repeats. Caveat that matters: the **leptomeningeal arm specifically was 16.7%**, far below the brain-met arms. So ALK is a coherent target; leptomeningeal benefit is not demonstrated.

- **Methotrexate / DHFR.** The strongest and most reproducible model result in the set — binding probability 0.992 in both repeats, affinity -2.063 / -2.219. This is also the least surprising: methotrexate-DHFR is one of the best-characterised drug-target pairs in medicine, so this mostly tells us the model behaves sensibly. Clinically it appears only inside multi-drug regimens here (NCT00801216, NCT00992602), so no single-drug conclusion can be drawn.

- **Adavosertib / WEE1.** Clean chemistry (co-crystal 5V5Y, affinity -2.125 / -2.219, probability 0.859 / 0.875, stable). But the trial is n=4 and reports only a maximum tolerated dose — there is no efficacy evidence at all.

Weak or discordant:

- **Tesevatinib / EGFR.** Strong model numbers (affinity -1.680 / -1.750, probability ~0.72) against a **0% response rate** in the leptomeningeal cohort of NCT02616393, with 12 of 20 patients dying. This is the clearest reminder in the whole report that good binding predictions do not translate into patient benefit. The structure evidence is also weaker — 1XKK does not contain tesevatinib.

- **Etoposide / topoisomerase II alpha.** Rank flipped between the two Boltz-2 repeats, so by the pipeline's own rule **no ordering is claimed**. Physically it is the worst-modelled case too: etoposide really acts on a protein-plus-DNA complex, and only the protein was modelled.

- **Sapanisertib / mTOR and capecitabine / thymidylate synthase.** Both dock into structures that do not contain the drug, and both score weakly (sapanisertib probability 0.47 / 0.44; capecitabine 0.41 / 0.38). Capecitabine is additionally a prodrug — the active species is its metabolite, so docking the parent molecule is the wrong question.

- **Telaglenastat, tucatinib, sorafenib, patupilone.** Middling binding probabilities (0.13-0.56) and trials of 2-22 patients. Nothing here supports a conclusion either way.

In every case the cousins MolMIM invented scored worse than the real drug (except the etoposide pair, where no ordering is claimed). That is the expected, reassuring result: these are optimised approved drugs and the cousins are untested machine edits. **No cousin in this run is a lead.**

6. Limitations and the real next step

- No control arms anywhere in the 19 trials; most enrolled single digits. Posted outcomes are descriptive only.

- Docking and affinity models score a static protein pocket. They do not model the blood-brain or blood-CSF barrier, which is the single biggest obstacle in leptomeningeal disease — a drug can bind perfectly and never reach the cerebrospinal fluid.

- Three drugs were docked into structures that do not contain them (sapanisertib, tesevatinib, capecitabine); one target (topoisomerase II alpha) was modelled without its DNA.

- Antibodies, radiolabelled antibodies, antibody-drug conjugates, cell therapy, radiation and the diagnostic device cannot be scored by these molecule models at all, and several of the 19 trials are exactly those.

- **The next step is not more compute.** It is a wet-lab assay: a binding or cell-viability measurement on the drug-target pairs flagged as coherent, and crucially a CSF-penetration measurement. No amount of additional modelling can substitute for that.

Model scores are hypotheses only — not clinical proof of efficacy or safety.